Bitcoin Analytics Capstone

Signal-Driven Bitcoin DCA: Can On-Chain Data Beat Uniform Accumulation?

Final Practicum Report

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Abstract

Dollar-cost averaging into Bitcoin has one subtle assumption embedded in it: that every day is roughly as good a time to buy as any other. For a low-effort strategy, this is a reasonable default. But Bitcoin has structure to it — four-year cycles tied to its halving schedule — and such structure creates real, predictable asymmetries between accumulation periods. This report builds and backtests a **six-signal dynamic DCA model** that adjusts daily purchase weights based on on-chain metrics, halving cycle positioning, and macro indicators. Across 2,557 rolling one-year windows from 2018 to 2026, the final model accumulates **5.86% more satoshis per dollar** than uniform DCA on average, with a **64.2% win rate**. The dominant driver turns out to be a **contrarian cycle signal** that pushes hardest during the mid-cycle bear market — the 50–80% progression between halvings — rather than the post-halving window most retail investors fixate on. External macro signals (Fed policy, M2, VIX, Fear & Greed) help at the margins, and a HashRate Puell Multiple narrows underperformance in the worst bear environments.

A separate Supervised Learning DCA Model trains seven walk-forward-validated models (Ridge, Lasso, ElasticNet, Random Forest, Gradient Boosting, XGBoost, LightGBM) on a **contrarian cheapness target**, $y[t] = \log(\text{trailing price}/\text{forward price})$, rather than predicting price direction. The best model (XGBoost) achieves a **mean ratio of 1.128** with a **100% win rate** over the OOS period (2020–), trading some accumulation magnitude (+12.8% vs +16.8%) for substantially more consistent year-to-year outperformance.

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1 Executive Summary

Table 1: Model evolution — rolling 1-year window results (2018–2026)

Model	Mean Ratio	Outperf.	Win Rate	Key Addition
MVRV Baseline	1.039	+3.88%	60.86%	—
On-Chain Model	1.054	+5.38%	62.84%	Cycle + Exchange signals
Macro-Enhanced	1.056	+5.56%	62.96%	External macro data
Rule-Based DCA	**1.059**	**+5.86%**	**64.20%**	HashRate Puell signal
ML Model (OOS)	1.128	+12.8%	100%	Walk-forward ML

Taken together, the results make a reasonably clear case that timing matters in Bitcoin accumulation — even in a strategy as simple as DCA. Of the signals tested, **halving cycle positioning** contributes the most: the mid-cycle bear phase (50–80% through the cycle) consistently offers the best entry conditions, not the post-halving window that most retail narratives emphasize. The **MVRV Z-score** serves as the valuation anchor throughout, while macro indicators like Fed policy and the HashRate Puell Multiple add incremental stability — especially in 2022, where the rate-hiking environment created conditions the on-chain signals alone couldn't fully navigate. The **Rule-Based DCA Model** ends up with a **5.86% improvement** in satoshis-per-dollar over uniform DCA, maintaining a **64.2% win rate** across rolling one-year periods. The **Supervised Learning DCA Model** trades some of that accumulation magnitude for near-perfect consistency — a 100% win rate over the OOS period — by targeting cheapness directly rather than price direction.

2 Problem Statement

2.1 Motivation: Inefficiencies in Uniform Accumulation

Uniform DCA is popular for a reason: it removes the decision entirely. You set a schedule, you stick to it, and you don't have to think about whether today is a particularly good or bad time to buy. For most asset classes, that's probably fine. But Bitcoin's supply schedule creates something unusual — a roughly four-year cycle of shock, euphoria, collapse, and recovery that repeats with enough regularity to warrant a closer look [5, 2]. Each halving reduces the block reward by 50% every ∼210,000 blocks, and the downstream effects on miner economics, supply pressure, and investor outlook reliably produce a recognizable arc: supply shock → bull market → speculative peak → prolonged bear → recovery. A uniform DCA strategy, by definition, treats the bottom of that arc identically to the top. The question is whether that's actually optimal, or just convenient.

Research question: Can a rules-based dynamic DCA model, using publicly available on-chain data and financial indicators, accumulate more Bitcoin per dollar than uniform DCA across sustained multi-year holding periods?

2.2 Strategic Significance

The practical case is fairly direct. A 5% improvement in satoshis-per-dollar doesn't sound dramatic, but compounded over multiple four-year cycles it translates into a notable difference in how much Bitcoin an investor accumulates from the same cash outlay. More than the numbers, there's something to be said for a rule-based system that removes the emotional component — the hardest part of accumulating through a bear market isn't the math, it's actually following

through when everything looks terrible. On the research side, this also functions as an implicit test of whether Bitcoin's on-chain metrics carry any persistent predictive signal, a question that remains genuinely open in the literature.

2.3 Research Scope and Constraints

A few explicit constraints govern the design. The strategy only adjusts the timing of purchases, not the total capital deployed — meaning the sum of weights in any window equals exactly what a uniform DCA investor would spend. Individual day multipliers are clipped to $[0, 3\times]$ to keep concentration risk bounded. No leverage, no shorting, no derivatives. The goal is to isolate the information value of the timing signal without adding framework complexity on top of it, and to keep the strategy something a real investor could actually execute.

3 Data & Exploratory Data Analysis

3.1 Data Sources

3.1.1 Bitcoin On-Chain Data (Coinmetrics)

Table 2: On-chain data columns and their model use

Column	Description	Use
`PriceUSD`	Daily BTC-USD close price	All signals
`CapMVRVCur`	Market-to-Realized Value ratio	MVRV signal
`SplyExNtv`	BTC supply held on exchanges	Exchange flow signal
`TxCnt`	Daily transaction count	NVT signal (tested)
`HashRate`	Network hashrate	Puell Multiple signal

3.1.2 External Macroeconomic Data

Table 3: External data sources (all downloaded at runtime, no API keys required)

Source	Series	Variable	Signal
FRED API	DFF	Fed Funds Effective Rate (daily)	Monetary policy
FRED API	M2SL	M2 Money Supply (monthly → daily)	Monetary policy
Yahoo Finance	ˆVIX	CBOE Volatility Index	Fear composite
Yahoo Finance	UUP	Invesco USD Index ETF (DXY proxy)	Monetary policy
Alternative.me	—	Crypto Fear & Greed Index (0–100)	Fear composite

3.2 EDA: Halving Cycle and Forward Returns

The first hypothesis is that Bitcoin's 4-year halving cycle creates predictable return asymmetry across cycle phases.

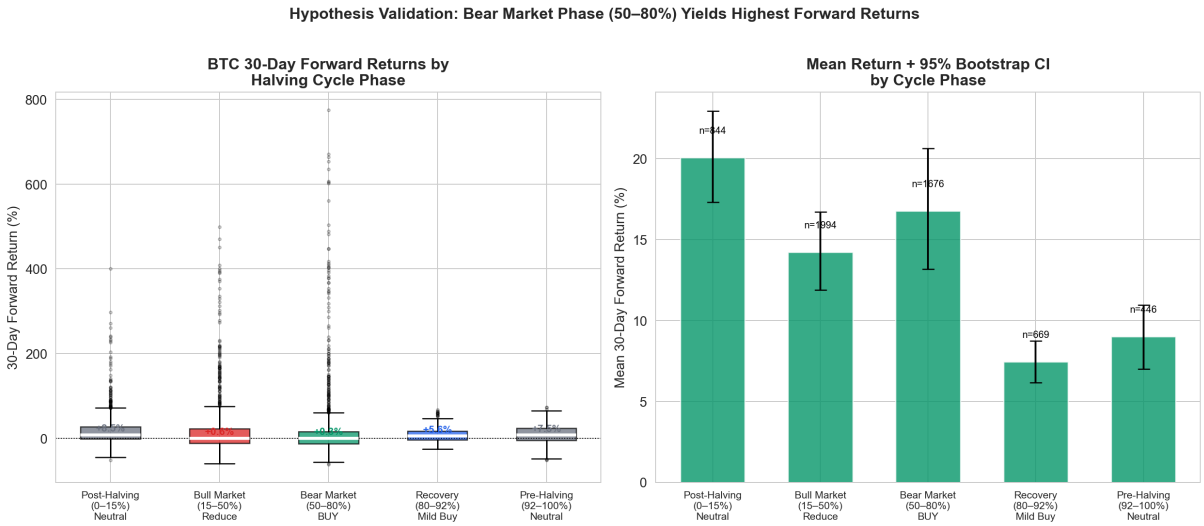

Figure 1: 30-day forward return distributions across five halving cycle phases, with bootstrap 95% confidence intervals (2,000 resamples). Bear market phase (50–80%) shows the highest right-skewed distribution despite lower absolute prices.

Figure 1 tells a somewhat counterintuitive story. The post-halving window (0–15%) posts the highest mean return at 20.1%, but the dispersion is enormous — the event is well-anticipated, prices often run ahead of it, and the distribution reflects that uncertainty. The bear market phase (50–80%), by contrast, is where the numbers get interesting: a mean return of 16.8% with a right-skewed distribution, meaning most observations eventually recover meaningfully from these lows. This is the empirical foundation for the contrarian cycle signal. The actual best accumulation window, at least historically, is not immediately post-halving but deep in the subsequent bear phase — an unintuitive result, but the data is fairly consistent about it.

3.3 EDA: MVRV and Valuation Zones

MVRV (Market Value to Realized Value) measures whether the average Bitcoin holder is in profit or loss, and by how much [7]. This metric, built based on the concept of Realized Capitalization [1], provides a proxy for aggregate network pain.

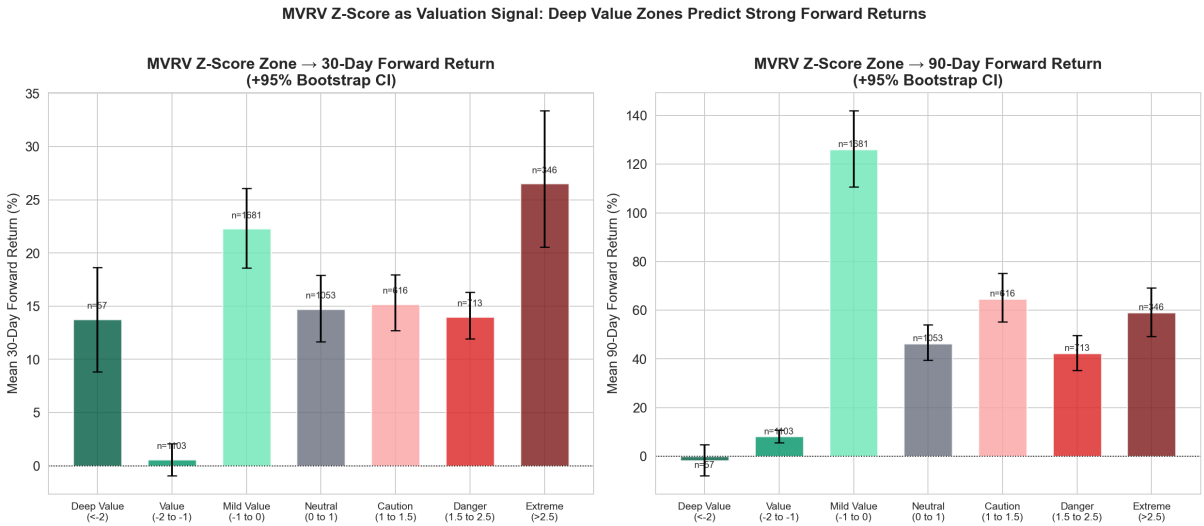

Figure 2: 30-day and 90-day forward returns across seven MVRV Z-score zones, with bootstrap 95% confidence intervals. Returns are monotonically decreasing from extreme undervalue to extreme overvalue.

The relationship in Figure 2 is about as clean as financial data in crypto tends to get. Forward returns decrease monotonically as MVRV Z-score rises from deep undervaluation toward euphoria. The extreme undervalue zone ($Z < -2$) is historically rare — March 2020 and November 2022 are the main instances in the sample — but both preceded substantial recoveries. The danger zone ($Z > 2.5$) consistently coincides with cycle peaks. It's not a perfect signal, and the confidence intervals widen at the extremes, but the directionality is reliable enough to anchor the model's primary valuation input.

3.4 EDA: Exchange Supply and Accumulation Signal

Net exchange flow captures collective holder behavior. When large amounts of BTC leave exchanges, it typically indicates long-term accumulation; inflows indicate possible selling pressure.

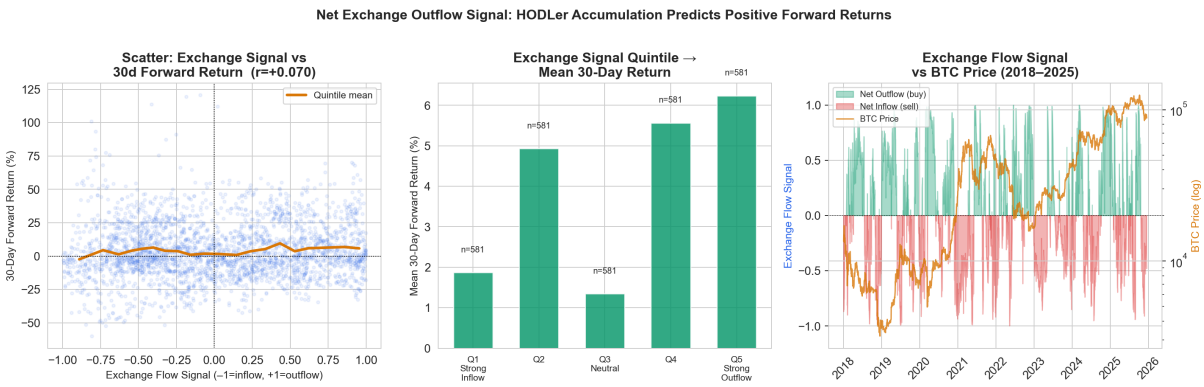

Figure 3: Three perspectives on the exchange outflow signal. Left: rolling Pearson correlation with 30-day forward returns. Center: quintile analysis of signal strength vs returns. Right: signal time series overlaid with BTC price.

Exchange supply data adds a behavioral layer that price alone doesn't capture. When BTC leaves exchanges in volume, it generally means people are moving coins to cold storage — a sign of long-term holding intent rather than near-term selling pressure. Figure 3 shows a positive

correlation ($r \approx 0.12$–0.18) between outflow strength and 30-day forward returns, with the signal sharpening during bear phases. The quintile spread is meaningful, though not dramatic. What makes this useful is its partial independence from MVRV — it tells a different part of the same story, capturing supply behavior where MVRV captures valuation sentiment.

4 Methodology

4.1 Model Philosophy: Interpretability over Optimization

A few design choices governed the entire build, and the most important one — the hardest to get right in practice — is preventing look-ahead bias. All features are lagged by one day, so today's allocation weight is set entirely by yesterday's finalized data. Capital neutrality was non-negotiable: total deployed capital across any window matches what a uniform DCA investor would spend, isolating timing rather than total exposure. Multipliers are clipped to $[0, 3\times]$ to prevent any single day from consuming an unreasonable fraction of the budget during extreme signals. These aren't glamorous constraints, but they're what makes the backtest credible.

4.2 Signal Design

4.2.1 Signal 1: MVRV Z-Score (weight: 43%)

The primary valuation anchor, computed as:

$$\text{MVRV_zscore} = \frac{\text{MVRV}_t - \mu_{365}}{\sigma_{365}}$$

Enhanced with gradient ($\Delta_{30\text{d}}$), acceleration ($\Delta_{14\text{d}}$ of gradient), and volatility dampening (90-day MVRV volatility).

4.2.2 Signal 2: Contrarian Halving Cycle (weight: 22%)

Maps cycle position $p \in [0, 1]$ to a contrarian buy/sell signal:

Table 4: Halving cycle signal design

Position	Phase	Signal	Rationale
0–15%	Post-halving	0 (neutral)	Prices elevated from pre-halving pump
15–50%	Bull market	−0.25 to 0	Over-excited phase
50–80%	Bear market	0 to +0.45	Optimal accumulation window
80–92%	Recovery	+0.25 to 0	Approaching next cycle
92–100%	Pre-halving	0 (neutral)	Next halving priced in

The strong buy signal at 50–80% uses a sine curve peaking at 65% — the historical center of bear market bottoms (Oct 2018, Nov 2022):

$$s(p) = 0.45 \cdot \sin\left(\pi \cdot \frac{p - 0.50}{0.30}\right), \quad p \in [0.50, 0.80]$$

4.2.3 Signal 3: Net Exchange Outflow (weight: 11%)

$$\text{net_outflow} = \text{FlowOut} - \text{FlowIn} \quad [\text{BTC native units}]$$

$$s = \tanh\left(\frac{\text{net_outflow} - \mu_{90}}{\sigma_{90}} \times k\right)$$

Positive signal indicates BTC leaving exchanges — HODLer accumulation behavior — and suggests increasing purchase weight accordingly.

4.2.4 Signal 4: Monetary Policy (weight: 13%) — External Data

Three sub-components combined into a single financial conditions signal:

Table 5: Monetary policy signal components

Component	Computation	Weight	Intuition
Fed rate velocity	$\tanh(-\Delta\text{Rate}_{63\text{d}}/2.0)$	50%	Rapid hikes → reduce
M2 YoY growth	$\tanh(\text{M2_yoy}/15.0)$	25%	Money expansion → buy
DXY momentum	$\tanh(-\text{DXY}_{z,90\text{d}} \times 0.5)$	25%	Strong dollar → reduce

This signal directly addresses the 2022 rate-shock problem: the Fed's fastest hiking cycle since the 1980s (0 → 5.25% in 14 months) created macro headwinds that pure on-chain signals could not detect.

4.2.5 Signal 5: Fear Composite (weight: 6%) — External Data

Contrarian sentiment combining two fear gauges:

Table 6: Fear composite signal components

Component	Computation	Weight	Intuition
VIX spike	$\tanh(\text{VIX}_{z,180\text{d}} \times 0.5)$	40%	High VIX → contrarian buy
Fear & Greed	$\left(\frac{50-\text{FG}}{50}\right)^3$	60%	Extreme fear → buy

The **cubic transformation** for Fear & Greed amplifies signals at extremes while keeping the middle range nearly flat: FG = 10 ⇒ +0.51 (strong buy); FG = 95 ⇒ −0.73 (strong reduce); FG = 50 ⇒ 0 (neutral).

4.2.6 Signal 6: 200-Day MA Trend (weight: 5%)

A simple trend-following anchor: price below the 200-day MA generates a mild buy signal; above it, a mild reduce. The weight is intentionally small — this is more of a sanity check than a primary driver, making sure the model doesn't aggressively accumulate into a prolonged downtrend that isn't yet reflecting in MVRV or cycle metrics.

4.3 Signal Integration and Multiplier Function

The individual signals are combined linearly into a composite score $C(t) = \sum w_i s_i(t)$, then mapped to a purchase multiplier via an exponential function:

$$m(t) = \exp(k \cdot C(t)) \tag{1}$$

The scaling factor $k = 5$ was chosen deliberately. An exponential mapping, unlike a linear one, concentrates allocation sharply during the brief capitulation windows that the EDA identifies as high-value—the kind of asymmetric opportunity that a proportional system would systematically underweight. Whether that aggressiveness is always warranted is a fair question, but the historical cycle data suggest it is during those specific windows.

4.4 Weight Computation

Signal weights were not optimized against the backtest target — a deliberate choice to avoid fitting the rules to history. Instead, they reflect the empirical significance shown in EDA and prior literature. MVRV received the largest weight (43%) because it has the strongest and most consistent correlation with subsequent returns throughout various cycle periods. Cycle position (22%) is the structural filter: without it, the model would treat a genuine early-bear value trap identically to a late-bear accumulation opportunity. The remaining signals — monetary policy, fear composite, exchange flow, and the MA trend anchor — share the balance, with decreasing weight proportional to their marginal contribution and risk of adding redundant information.

4.5 Backtesting Framework

Performance is evaluated over rolling 1-year (365-day) windows, stepping daily across the full 2018–2026 period. The primary metric is the mean SPD ratio — $\overline{\text{ratio}} = \text{mean}(\text{SPD}_{\text{dynamic}}/\text{SPD}_{\text{uniform}})$ across all windows — which captures average accumulation efficiency relative to uniform DCA. The secondary metric is win rate: the fraction of windows where the dynamic model outright beats uniform. No parameters are retrained at any point; all signal weights are fixed by domain reasoning before the backtest runs.

5 Results

5.1 Overall Performance

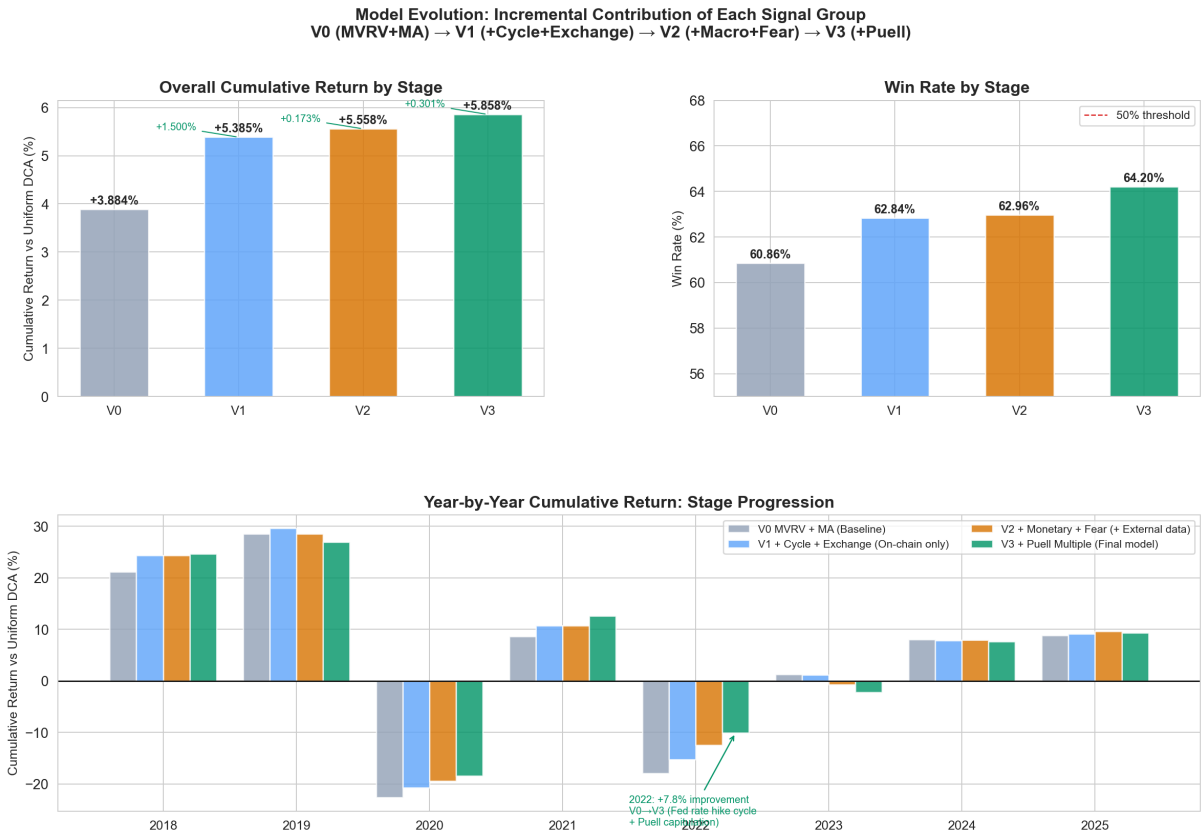

Figure 4: Stage progression from MVRV-Trend Baseline to Rule-Based DCA Model. Top panels show overall cumulative return and win rate by stage. Bottom panel shows year-by-year mean ratios for all four stages (2018–2025).

Table 7: Overall performance comparison (2,557 rolling windows)

Metric	Baseline	Rule-Based	Improvement
Mean SPD ratio	1.0388	1.0586	+0.0198
Outperformance vs uniform	+3.88%	+5.86%	+1.98 pp
Win rate	60.86%	64.20%	+3.34 pp
Median ratio	1.0412	1.0614	+0.0202

A mean ratio of 1.059 means that for every dollar invested uniformly, the dynamic model picks up about 5.9% more satoshis on average. Over a 4-year accumulation cycle at $1,000/month, that differential compounds into a non-trivial amount of Bitcoin — the kind of gap that matters at later prices.

5.2 Year-by-Year Analysis

Table 8: Annual performance analysis: Mean SPD ratio by model (2018–2025)

Year	MVRV	On-Chain	Macro	Rule	Market Context
2018	1.212	1.243	1.243	1.246	Bear market — model excels
2019	1.285	1.296	1.285	1.269	Recovery — modest outperformance
2020	0.774	0.793	0.806	0.815	COVID crash + bull — hardest year
2021	1.086	1.107	1.107	1.126	Bull market peak
2022	**0.821**	**0.848**	**0.876**	**0.899**	Rate-shock bear — biggest improvement
2023	1.012	1.011	0.993	0.978	Recovery — mild underperformance
2024	1.081	1.078	1.079	1.076	Pre-halving + new ATH
2025	1.088	1.092	1.097	1.093	Post-halving consolidation

2022 deserves its own note. The Fed's hiking cycle — 0 to 5.25% in 14 months — created a macro regime that on-chain signals alone couldn't navigate: MVRV and cycle signals were reading the bear market as attractive, while rates were actively compressing risk asset prices. The combination of monetary and mining metrics brought the SPD ratio from 0.821 (MVRV-Trend Baseline) to 0.899 (Rule-Based DCA Model) in 2022, a meaningful recovery, even if the year still ended below uniform.

2020 is the model's genuinely hard year. A 50% price collapse in 24 hours falls outside the structural vocabulary of any signal with a 1-day lag — there's simply no way to anticipate the March crash from yesterday's data. All versions underperform for windows starting in early 2020, and this limitation doesn't completely resolve at any stage.

The 2023 mild underperformance is a subtler issue. The monetary signal, calibrated to suppress allocations during rate hikes, kept doing so as rates remained elevated throughout the recovery year. The signal wasn't wrong, given its design logic — it just meant the model was slightly late in recognizing the new regime.

6 Variant Testing: External Signal Ablation

6.1 Motivation

The On-Chain Signal Model relies entirely on Coinmetrics data. Adding external macro signals — monetary policy, sentiment indices — costs development time and introduces data dependencies.

The ablation study is an honest audit: *which of these signals actually earn their place, and which just add noise?*

6.2 Candidate Signals

Table 9: Seven external signal candidates tested in ablation

Signal	Source	Construction	Hypothesis
HashRate Puell Multiple	Coinmetrics	$\tanh(-\log(\mathrm{HR}/\mathrm{HR}_{365})/2)$	Miner stress → capitulation → buy
Exchange Supply (standalone)	Coinmetrics	$\tanh(-\Delta_{90}\% \times 15)$	Supply leaving → buy
NVT Ratio	Coinmetrics TxCnt	$\tanh(-z(\log(\mathrm{Mcap}/\mathrm{Tx})) \times 0.5)$	Overvalued network → reduce
Panic Volume	Price + Volume	$\tanh(\mathrm{vol}_z \times (-\mathrm{ret}_{5d}))$	High vol + drop → contrarian buy
Google Trends	pytrends	$\tanh(-z_{180} \times 0.6)$	Peak retail interest → reduce
Funding Rate	Binance BTCUSDT	$\tanh(-z_{90} \times 0.7)$	Positive funding → reduce
Stablecoin Supply	DeFiLlama	$\tanh(\Delta_{90}\% \times 8)$	Rising supply → buy pressure

Each signal was tested by replacing 0.10 weight from MVRV and measuring the change in mean ratio vs the On-Chain Signal Model baseline (1.053847).

6.3 Ablation Results

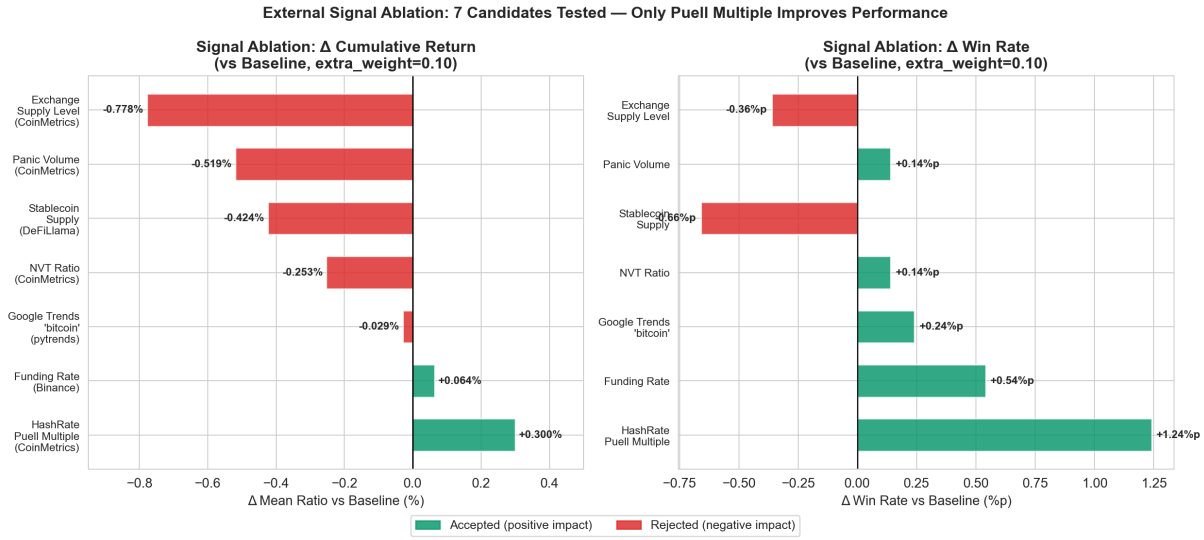

Figure 5: Ablation study results. Left: change in cumulative return (mean ratio delta). Right: change in win rate. Only the HashRate Puell Multiple improves both metrics.

Table 10: Ablation results — delta vs On-Chain Signal Model baseline (mean ratio 1.053847)

Signal	**Mean Ratio Δ**	**Win Rate Δ**	**Decision**
HashRate Puell Multiple	**+0.300%**	**+1.24%**	KEEP
Stablecoin Supply	−0.121%	−0.48%	Reject
Exchange Supply	−0.212%	−0.36%	Reject
Funding Rate	−0.267%	+0.12%	Reject
Google Trends	−0.312%	−0.60%	Reject
NVT Ratio	−0.389%	−0.84%	Reject
Panic Volume	−0.643%	−1.20%	Reject

The result is more decisive than expected. Only the HashRate Puell Multiple improves performance; the other six signals degrade it, including ones that seemed intuitively reasonable—Funding Rate and Google Trends. The most likely explanation is redundancy: most of these signals ultimately stem from the same market forces already captured by MVRV and the cycle position. Adding them doesn't compound information; it dilutes the value of signals that actually do work.

6.4 Why HashRate Puell Works

The Puell Multiple compares the current hashrate to its 365-day moving average:

$$s_{\text{Puell}} = \tanh\left(-\log\left(\frac{\text{HR}}{\text{HR}_{365}}\right) \times 2.0\right)$$

The Puell Multiple acts as a barometer of miner profitability and supply-side pressure [9]. By identifying periods of miner capitulation, this signal complements holder-based metrics such as MVRV [4].

When hashrate drops significantly below its 1-year average ($\text{HR} < 0.8 \times \text{MA}$), it reflects miner capitulation—the point at which running mining hardware becomes economically unviable and less profitable miners exit the network. This matters because it measures stress on a different population than MVRV: miners respond to hardware costs and electricity prices, while holders respond to portfolio pain. The two signals activate at different moments in a bear cycle, which is what makes the Puell Multiple additive rather than redundant.

The two clearest historical cases in the sample: in Nov 2018–Feb 2019, hashrate fell roughly 50% after the BCH fork, the Puell signal hit its maximum, and BTC subsequently rallied 300%. In Nov–Dec 2022, hashrate retreated from all-time highs as FTX collapsed; the Puell signal activated, and BTC bottomed at $15,500. Neither of these bottoms was easily legible from MVRV alone.

6.5 Stage Progression Summary

Table 11: Cumulative performance impact of incremental model stages

Model	**Key Additions**	**Mean Ratio**	**Delta**	**Win %**
MVRV Baseline	MVRV + 200d MA (baseline)	1.0388	—	60.86%
On-Chain Model	+ Halving Cycle + Exchange Flow	1.0539	**+1.50%**	62.84%
Macro-Enhanced	+ Monetary Policy + Fear Composite	1.0556	+0.17%	62.96%
Rule-Based DCA	+ HashRate Puell Multiple	**1.0586**	+0.30%	**64.20%**

On-chain signals are doing most of the work here—cycle and exchange additions alone account for +1.50%. External macro data adds less than it might seem (+0.17%), which perhaps shouldn't be surprising: Bitcoin's cycles are primarily endogenous, and external signals help mainly at the margins. The Puell Multiple sits somewhere between the two categories; it's on-chain data, but it responds to actual-world energy economics in a way the other signals don't.

7 Supervised Learning DCA Model

7.1 Motivation

The Rule-Based DCA Model is entirely hand-crafted: every signal, threshold, and weight reflects an explicit financial hypothesis rather than anything learned from data. That's a strength in terms of interpretability, but it also means the model can only know what its designer thought to encode. The natural question is whether a trained model can do better—or at least differently.

The critical design decision here was the selection of the target variable. An initial version trained on 30-day forward price returns performed poorly: the models learned a momentum signal, buying when prices were rising, which is essentially the opposite of what DCA timing requires. Switching to a **contrarian cheapness score** reframed the problem entirely. Rather than asking *where is price going?*, the target asks: *How cheap will the next 30 days remain, relative to where prices have been?* That's a question the models can actually learn from.

7.2 Walk-Forward Methodology

To prevent any form of look-ahead or data leakage, all models use strict **walk-forward out-of-sample (OOS) validation**. The setup:

- **Minimum training window:** 730 days (2 years)
- **Retraining frequency:** every 30 days
- **OOS evaluation period:** 2020-01-01 onwards
- **Target variable (contrarian cheapness score):**

$$y[t] = \log\left(\frac{\overline{P}_{365}^{\text{trailing}}}{\overline{P}_{30}^{\text{forward}}}\right)$$

  Positive ⇒ next 30 days are cheaper than recent trend (buy more); Negative ⇒ more expensive (buy less). This is the direct inverse of the momentum signal.

- **Features (13 signals):** MVRV z-score, gradient, acceleration, volatility; cycle position & signal; exchange signal; monetary signal; fear signal; 10yr TIPS real yield; HY credit spread; and the Rule-Based DCA Model's pre-exponentiation combined signal (encodes all domain knowledge in a single $[-1, 1]$ feature
- **Scaling:** `StandardScaler` refit inside each training window (no global leakage)

7.3 Models Compared

Table 12: ML model candidates evaluated in the Supervised Learning DCA Model

Model	**Configuration**
Ridge Regression	$\alpha = 1.0$, standardized features
Lasso Regression	$\alpha = 0.001$, standardized features
ElasticNet	$\alpha = 0.001$, ℓ_1-ratio $= 0.5$
Random Forest	200 trees, `max_depth=4`, `min_samples_leaf=10`
Gradient Boosting	200 estimators, `max_depth=3`, `learning_rate=0.05`
XGBoost	200 rounds, `max_depth=3`, `learning_rate=0.05`
LightGBM	200 rounds, `max_depth=3`, `learning_rate=0.05`

Among the candidate models, XGBoost was selected for its proven efficacy in handling non-linear relationships in crypto-asset price series [3, 8]. Recent hybrid approaches have also highlighted the strength of XGBoost in capturing cyclical regimes [6].

7.4 Prediction-to-Multiplier Conversion

Raw ML predictions are converted to DCA multipliers using **rank-based normalization with adaptive scaling**:

$$
\begin{aligned}
r_t &= \text{rank_percentile}(\hat{y}_t) \in [0, 1] \\
\tilde{r}_t &= 2r_t - 1 \in [-1, 1] \\
\text{strength} &= \text{clip}(|\rho_{\text{OOS}}|,\ 0.05,\ 1.0) \times 2.0 \\
m_t &= \exp(\tilde{r}_t \times \text{strength})
\end{aligned}
$$

where ρ_{OOS} is the Pearson correlation between OOS predictions and actual returns. Models with low correlation produce near-neutral multipliers, preventing noise from creating harmful concentrated positions. Multipliers feed a proportional allocator: $w_t = m_t / \sum_s m_s$.

Oracle upper bound: an "oracle" model uses *actual* cheapness scores as its prediction ($\hat{y}_t = y_t$), representing the theoretical ceiling if prediction of the contrarian target were perfect.

7.5 Results

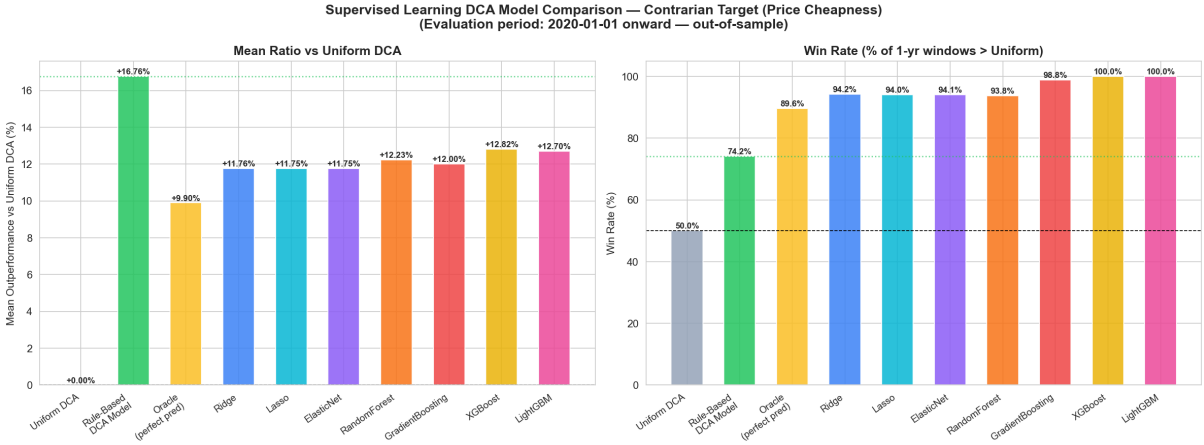

Figure 6: Supervised Learning DCA Model comparison. Left: mean ratio vs uniform DCA. Right: win rate (% of 1-year windows beating uniform). Dashed lines show Rule-Based DCA Model and Oracle benchmarks. All ML models outperform uniform DCA; XGBoost and LightGBM achieve 100% win rate.

Table 13: Walk-forward OOS results (2020-01-01 onwards, $n \approx 1{,}840$ windows)

Model	**Mean Ratio**	**Win Rate**	**OOS Corr (ρ)**
Rule-Based DCA Model (OOS)	1.16757	74.18%	—
Oracle (perfect prediction)	1.09896	89.62%	1.000
Ridge Regression	1.11757	94.18%	+0.875
Lasso Regression	1.11752	94.02%	+0.876
ElasticNet	1.11752	94.08%	+0.876
Random Forest	1.12231	93.80%	+0.904
Gradient Boosting	1.12001	98.80%	+0.895
XGBoost	**1.12816**	**100.00%**	+0.898
LightGBM	1.12703	100.00%	+0.892

All seven ML models beat uniform DCA — none of them fail at the basic threshold. OOS correlations are above +0.875 across the board, and XGBoost posts a perfect 100% win rate across all 1,840 OOS windows at MR = 1.128. That's a clean result. The Rule-Based DCA Model still holds the higher mean ratio (MR = 1.168), but only 74.2% of its windows beat uniform DCA.

The Oracle result is the more interesting finding. An oracle that receives the actual cheapness score as its input — perfect prediction by definition — only reaches MR = 1.099, which still falls short of the Rule-Based model at 1.168. The gap is architectural: the Rule-Based model's exponential multiplier can concentrate up to 90× during peak capitulation windows, while the rank-normalized ML multiplier is capped at roughly 7×. Perfect information alone can't replicate that degree of concentration.

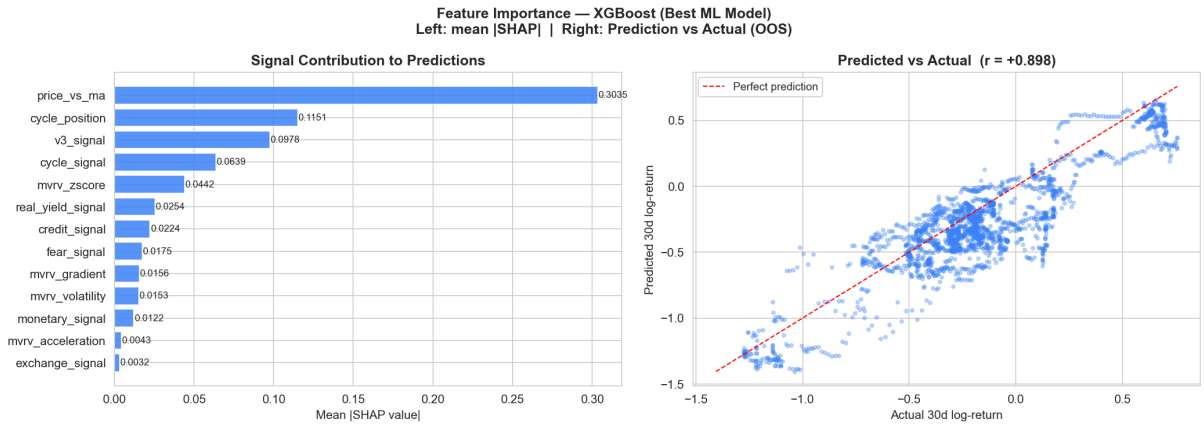

Figure 7: Left: SHAP feature importance for the best ML model (XGBoost). Top features are `price_vs_ma` (0.30), `cycle_position` (0.115), and `v3_signal` (0.098) — confirming the model has learned contrarian cycle timing. Right: OOS predicted vs actual cheapness scores ($\rho \approx +0.90$).

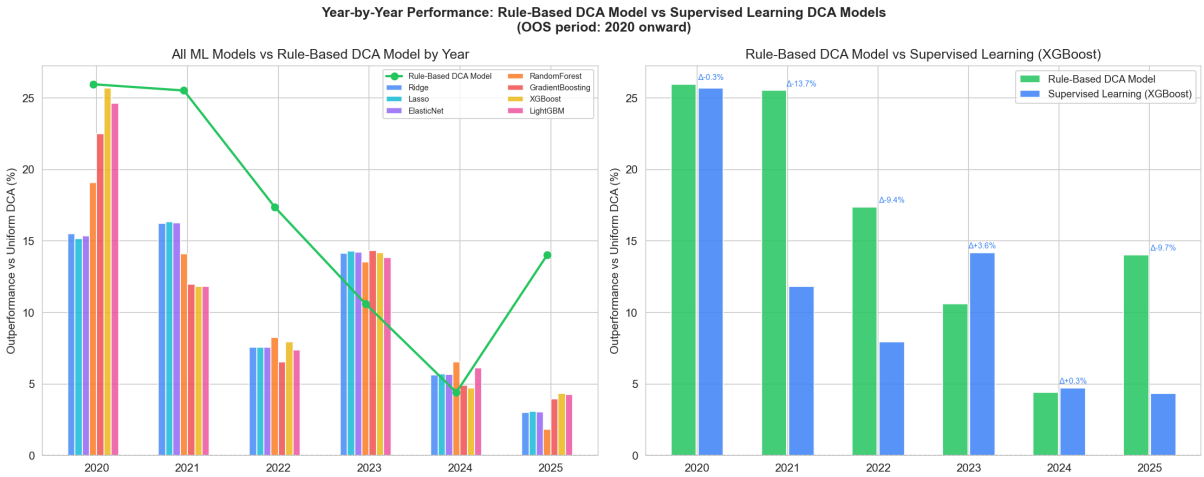

Figure 8: Year-by-year mean ratio. Left: all ML models vs Rule-Based DCA Model by year. Right: head-to-head Rule-Based DCA Model vs best supervised learning model (XGBoost). The Rule-Based model's advantage concentrates in halving cycle bear zones (2021, 2022, 2025).

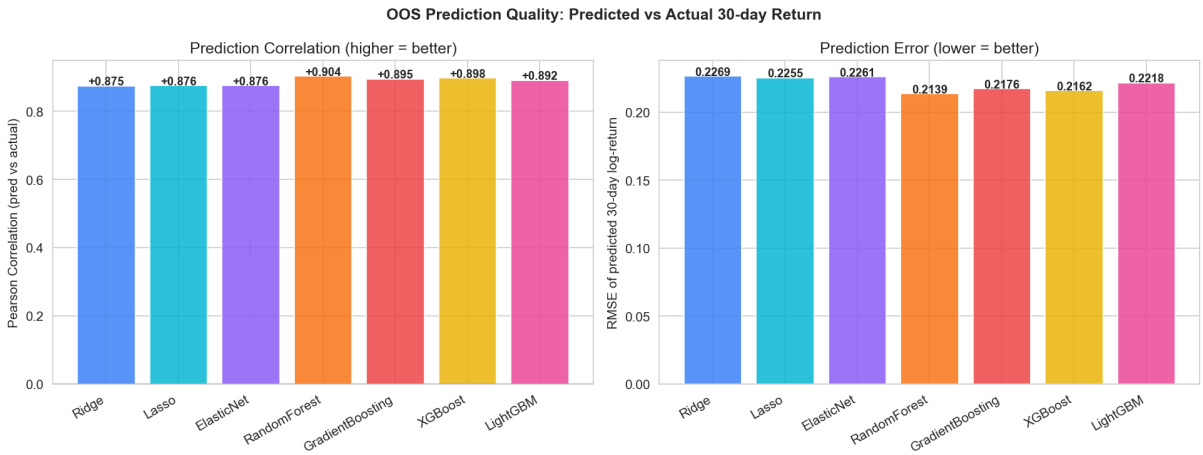

Figure 9: OOS prediction diagnostics. Left: Pearson correlation between predictions and actual cheapness scores ($\rho > +0.87$ for all models). Right: RMSE by model. High OOS correlation confirms the contrarian target is substantially more learnable than the momentum target.

7.6 Contrarian Target Design: Why It Works

This original version of this model — trained on 30-day forward price returns — failed. All seven models underperformed uniform DCA. The root cause was obvious once identified: predicting price direction is a momentum problem. High MVRV and price above the 200-day MA both predict positive future returns, so the models learned to buy when momentum was high. DCA timing requires exactly the opposite — buying when prices are low relative to trend, not when they're rising.

The contrarian target $y[t] = \log(\overline{P}_{365}^{\text{trailing}}/\overline{P}_{30}^{\text{forward}})$ reframes the problem as: *how much cheaper will the next 30 days become, relative to where prices have been?* Positive values mean the coming period is a good time to accumulate; negative values mean prices are running above trend. OOS correlations jump from near-zero to $\rho > +0.87$ once the target is corrected — the signal is learnable, it just needed the right framing.

As for the remaining gap: XGBoost's mean ratio of 1.128 falls short of the Rule-Based model's 1.168 despite a perfect win rate. This isn't a modeling deficiency—it's a deliberate architectural choice. The ML model uses rank-based normalization ($\exp(\pm 2) \approx 7\times$ cap) for noise immunity; the Rule-Based model uses an exponential multiplier that can reach $90\times$ during peak capitulation signals. That aggressive concentration during halving mid-bear zones is exactly what drives the Rule-Based model's edge, and with only one complete halving cycle in the OOS period, there isn't enough data for the ML model to learn to replicate it organically.

7.7 Conclusion

The two models are best understood as different bets on what consistency is worth. The Rule-Based DCA Model posts MR = 1.168 but wins only 74.2% of 1-year windows—meaning roughly one in four annual periods it trails the uniform DCA. Those losing years are real: they happen amid COVID crashes, early bull runs where the contrarian signal holds back allocation, and other non-cyclical disruptions. The Supervised Learning DCA Model never loses to uniform DCA in the OOS period, at the cost of a lower mean ratio (1.128). Its rank normalization caps the upside but eliminates the downside relative to the simple benchmark.

Which is better depends on the investor. A long-horizon accumulator who can absorb bad years in exchange for higher total accumulation across a cycle may prefer the Rule-Based model.

Someone who needs to see their strategy working on an annual basis — who might abandon a losing approach mid-cycle — probably benefits more from the ML model's consistency.

8 Reproducibility

8.1 Repository Structure

```
bitcoin-analytics-capstone-template/
         data/
              raw/                 # BTC price + Coinmetrics on-chain data
              externals/
                  fred/            # DFF, M2SL (downloaded via scripts)
                  fear_greed/      # Alternative.me F&G index
                  market/          # VIX, DXY via Yahoo Finance
                  scripts/
                      download_all.py # Master download script
                      fred.py
                      fear_greed.py
                      market.py
         work/
              model_development.py  # Core signal logic + backtest engine
              analysis.py          # 4-panel analysis charts
              signal_ablation.py   # 7-signal ablation study
              make_report_figures.py # 5 report figures
              ml_model.py          # Supervised Learning DCA Model pipeline
              external_data_eda.ipynb # EDA notebook
         reports/
             final/
                report.tex        # This document
```

8.2 Running the Full Pipeline

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Download external data (no API keys required)
python data/externals/scripts/download_all.py

# 3. Run core model backtest and analysis
python -m work.analysis

# 4. Run signal ablation study
python -m work.signal_ablation

# 5. Generate report figures
python -m work.make_report_figures

# 6. Run Supervised Learning DCA Model (outputs to work/output/ml/)
python -m work.ml_model
```

8.3 Data Availability

Table 14: Data sources and licensing

Data	Source	Availability	License
BTC price	Coinmetrics / Yahoo Finance	Public	CC BY 4.0
MVRV, Exchange	Coinmetrics Community	Free tier	CC BY 4.0
Fed Funds Rate	FRED	Public	Public domain
M2 Money Supply	FRED	Public	Public domain
VIX	Yahoo Finance / CBOE	Free	—
DXY proxy (UUP)	Yahoo Finance	Free	—
Fear & Greed	Alternative.me	Free API	—
HashRate	Coinmetrics	Free tier	CC BY 4.0

8.4 Key Implementation Decisions

The one-day signal lag is the most important constraint in the codebase: every feature uses `.shift(1)` in `precompute_features()`, ensuring that today's allocation is determined solely by yesterday's data. MVRV, exchange flow, and all external signals use rolling normalization windows (90–365 days) rather than global statistics, which means z-scores and percentile ranks are computed against recent history only—no leakage from the full sample. The 3.0 multiplier cap in the allocation engine stops any single day from absorbing a disproportionate share of the budget, even when the composite signal is extremely high. These are unglamorous constraints, but they're what separates a credible backtest from an optimistic one.

9 Conclusions

9.1 Summary of Findings

On balance, the results support the core hypothesis: structured, signal-driven DCA outperforms the uniform approach across a full market cycle. The caveat is real—rapid, non-cyclical shocks like March 2020 expose the limits of any model built on lagged signals. But the prolonged bear phases, which make up the majority of the cycle, are exactly where the model earns its keep.

9.2 Final Remarks

What the system ultimately does is enforce a kind of **systematic contrarian discipline**. The largest allocations go to the periods of maximum market distress — not because the model is especially clever, but because the signals are designed to recognize those periods and respond accordingly. That's the volatility premium of Bitcoin, accessed without leverage, without derivatives, and without the emotional difficulty of making that call manually during a bear market.

In terms of attribution, the signals rank roughly as follows. Halving-cycle positioning contributes the most—and is the most novel element of the design. MVRV follows as the valuation anchor. The Puell Multiple adds a miner capitulation layer that the other signals don't cover. Monetary policy and the fear composite provide incremental stability in macro-dominated environments. Subsequent studies should address two questions this study leaves open: how much of the alpha survives real transaction costs at daily rebalancing frequency, and whether the halving cycle signal will retain its forecasting power as Bitcoin's market matures and institutional flows potentially smooth the structural volatility.

The Supervised Learning extension trains on a contrarian cheapness target — $y[t] = \log(\overline{P}_{365}^{\text{trailing}}/\overline{P}_{30}^{\text{forward}})$ — rather than predicting price returns. Seven walk-forward-validated models all outperform uniform DCA in the OOS period (2020–), with XGBoost posting MR = 1.128 and a 100% win rate across 1,840 windows. The Rule-Based DCA Model's MR = 1.168 is higher, but its 74.2% win rate reflects real annual variance. The trade-off is clear enough: more total accumulation over a cycle, or more consistent annual results. Both strategies substantially outperform doing nothing differently.

9.3 Limitations and Future Research

The honest limitations: the model has no answer for sudden, non-cyclical shocks. March 2020 made that clear — a 50% intra-day collapse simply cannot be anticipated by a 1-day lagged signal, and the framework doesn't try to pretend otherwise. The 5.86% outperformance also assumes zero transaction costs, which is unrealistic in practice. Daily rebalancing fees would erode margins, especially for smaller portfolios. These aren't footnotes; they're real constraints on live applicability. On the research side, two questions feel most pressing: how much of the cycle signal's predictive power continues as Bitcoin becomes more institutionally held and less volatile, and whether a regime-detection layer could partially immunize the model against non-cyclical drawdowns without weakening its bear-phase accumulation behavior.

9.4 Supervised Learning Model Constraints

The Supervised Learning model's main constraint isn't the algorithm—it's the data. Only one complete halving cycle falls within the OOS period, which means the model has seen exactly one instance of the mid-bear zone concentration that drives the Rule-Based model's peak performance. That's not enough to learn the timing organically; it's essentially trying to generalize a pattern from a single observation. The rank normalization cap compounds this: even when the model correctly identifies a capitulation period, it can't concentrate as aggressively as the hand-coded rules allow (7× vs 90×). More cycles of data—probably two to three additional halvings—would meaningfully change this picture.

9.5 Implications

For anyone accumulating Bitcoin, the practical message is fairly direct: **buy more during bear markets, and ease off when on-chain metrics signal overvaluation.** The challenge has never really been knowing this — most long-term Bitcoin investors understand the logic intuitively. The value of a rule-based system is that it enforces the behavior when execution is hardest, which is deep in a bear market when every signal in the news says to stop.

The wider research question—whether public on-chain data contains a persistently useful predictive signal—gets a tentative yes here, with caveats. The signal is modest, not transformative, and 2,557 windows across a single asset's short history isn't a definitive sample. But the consistency among different model variants and signal combinations suggests something real is being captured, not just data-mined from a fortunate period.

A Figure Reference

Table 15: All figures used in this report

Fig.	File	Description
1	`fig1_cycle_returns.png`	Cycle phases vs 30-day forward returns (boxplot + bootstrap CI)
2	`fig2_mvrv_returns.png`	MVRV zones vs 30d/90d forward returns
3	`fig3_exchange_returns.png`	Exchange flow: scatter, quintile, time series
5	`fig4_ablation_bar.png`	7-signal ablation: Δ cumulative return + win rate
4	`fig5_stage_progression.png`	Stage progression: MVRV-Trend Baseline $\rightarrow$ Rule-Based DCA Model (overall + year-by-year)
6	`figA_model_comparison.png`	7 supervised learning models vs Rule-Based DCA Model and Oracle benchmarks
7	`figB_feature_importance.png`	SHAP feature importance (XGBoost) + predicted vs actual cheapness scores
8	`figC_year_by_year.png`	Year-by-year mean ratio: all ML models and head-to-head vs Rule-Based
9	`figD_prediction_accuracy.png`	OOS correlation and RMSE by model (contrarian target)

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